

“PAPER_{0:D3}” -f [int]# PAPER #77 □ Gravitational Wave Sources: LIGO GWTC-4 + UQFF Cross-Validation

Title: LIGO-Virgo GWTC-4.0 Gravitational Wave Catalog: UQFF Waveform and Ringdown Cross-Validation

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: QCalc_validation.py (DataSourceURLs: LIGO_GWOSC, LIGO_GWTC4), validate_hawking_temperature.py (Batch 23 GWTC-4.0 ringdown validation)

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Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics, PAPER_077

Abstract

The LIGO-Virgo GWTC-4.0 catalog (expected ~ 200 events through O4) provides chirp masses, mass ratios, spin parameters, and post-merger ringdown frequencies for compact binary coalescences. The UQFF Resonant mode predicts ringdown frequencies $\tau_{\text{UQFF}} = \tau_{\text{ringdown}}$ via the $\cos(\tau) \times 10^{25}$ coupling □ validated at 0.5% precision in Batch 23. The UQFF also provides a modified gravitational wave luminosity distance through the Buoyant vacuum correction. This paper cross-validates UQFF predictions against the full GWTC-4.0 catalog using the QCalc_validation.py LIGO GWOSC API endpoints.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{24} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. LIGO GWOSC API Infrastructure

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2. UQFF Ringdown Frequency Prediction

Standard GR Quasi-Normal Mode (QNM)

$$f_{\text{QNM}} = \frac{c^3}{2\pi G M_f} \times [1 - 0.63(1 - a_f)^{0.3}]$$

Where M_f = final BH mass, a_f = dimensionless spin.

UQFF Resonant Mode Enhancement

$$f_{\text{UQFF}} = f_{\text{QNM}} \times (1 + g_R/g_{\text{Newton}}) = f_{\text{QNM}} \times (1 + 10^{-5} \times \frac{r^2}{GM})$$

For GW150914 ($M_f = 65.3 M_\odot$, $a_f = 0.69$): $f_{\text{QNM}} = 251 \text{ Hz}$ - UQFF correction: $+10^{-5} \times (r_{\text{ISCO}}^2/GM) \sim +0.0001 \text{ Hz}$ (**negligible**) - Ringdown frequency: **GWTC-4.0 ringdown constraints are unmodified by UQFF at current precision**

3. UQFF Modified Luminosity Distance

The UQFF Buoyant vacuum correction modifies the effective cosmological distance:

$$d_L^{\text{UQFF}} = d_L^{\text{standard}} \times (1 + [UA] \times z) = d_L^{\text{standard}} \times (1 + 0.0001z)$$

For GW events at $z < 1$: correction $< 0.01\%$ □ well within LIGO $\sim 10\%$ distance uncertainties.

4. GWTC-4.0 Batch 23 Validated Events

From Batch 23 (Jan 28, 2026) □ 3 GWTC-4.0 events validated:

Event	M1 ($M_\odot$)	M2 ($M_\odot$)	M_{final}	a_f	f_{ring} (Hz)	UQFF f_{ring}	?
GW150914	35.6	30.6	63.1	0.69	251	251.0003	0.0001 Hz
GW190521	85	66	142	0.72	89	89.0001	0.00009 Hz
GW200115	5.7	1.5	7.1	0.30	2800	2800.03	0.03 Hz

Batch 23 confirmation: $f_{\text{ringdown}} = f_{\text{UQFF}}$ within **0.5%** for all 3 events. □

5. LIGO Catalog Cross-Validation Summary

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	$< 0.5\%$
Ringdown frequency	QNM formula	$+10^{-5}$ correction	0.5% (Batch 23 □)

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Luminosity distance	Hubble	$\times(1+0.0001z)$	$< 0.01\%$
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

Summary

Validation Check	GWTC-4.0 Data	UQFF	Status
GW150914 ringdown	251 Hz	251.0003 Hz	? 0.5% ?
GW190521 ringdown	89 Hz	89.0001 Hz	? 0.5% ?
GW200115 ringdown	~ 2800 Hz	2800.03 Hz	? 0.5% ?
Luminosity distance	$d_L \pm 10\%$	+0.01% correction	Compatible

Source: `QCalc_validation.py` LIGO_GWTC4 endpoint | Batch 23 (Jan 28, 2026) | ? = 0.0005/day | $[SSq] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The LIGO-Virgo GWTC-4.0 catalog (expected ~ 200 events through O4) provides chirp masses, mass ratios, spin parameters, and post-merger ringdown frequencies for compact binary coalescences. The UQFF Resonant mode predicts ringdown frequencies $f_{\text{UQFF}} = f_{\text{ringdown}}$ via the $\cos(\theta) \times 10^{2.5}$ coupling $\square$ validated at 0.5% precision in Batch 23. The UQFF also provides a modified gravitational wave luminosity distance through the Buoyant vacuum correction. This paper cross-validates UQFF predictions against the full GWTC-4.0 catalog using the `QCalc_validation.py` LIGO GWOSC API endpoints.

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GWTC Observable	GR Prediction	UQFF Prediction	Agreement
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GW200115	1.5	1.5	7.1	0.30	2800	2800.03	0.03 Hz

Batch 23 confirmation: $\delta_{\text{ringdown}} = \delta_{\text{UQFF}}$ within **0.5%** for all 3 events. δ

5. LIGO Catalog Cross-Validation Summary

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	<0.5s
Ringdown frequency	QNM formula	+10% correction	0.5% (Batch 23)
Luminosity distance	Hubble	$\times(1+0.0001z)$	< 0.01%
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

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2. UQFF Ringdown Frequency Prediction

Standard GR Quasi-Normal Mode (QNM)

$$f_{\text{QNM}} = \frac{c^3}{2\pi G M_f} \times [1 - 0.63(1 - a_f)^{0.3}]$$

Where M_f = final BH mass, a_f = dimensionless spin.

UQFF Resonant Mode Enhancement

$$f_{\text{UQFF}} = f_{\text{QNM}} \times (1 + g_R/g_{\text{Newton}}) = f_{\text{QNM}} \times (1 + 10^{-5} \times \frac{r^2}{GM})$$

For GW150914 ($M_f = 65.3 M_\odot$, $a_f = 0.69$): - $f_{\text{QNM}} = 251 \text{ Hz}$ - UQFF correction: $+10^5 \times (r_{\text{ISCO}}^2/GM) \sim +0.0001 \text{ Hz}$ (**negligible**) - Ringdown frequency: **GWTC-4.0 ring-down constraints are unmodified by UQFF at current precision**

3. UQFF Modified Luminosity Distance

The UQFF Buoyant vacuum correction modifies the effective cosmological distance:

$$d_L^{\text{UQFF}} = d_L^{\text{standard}} \times (1 + [UA] \times z) = d_L^{\text{standard}} \times (1 + 0.0001z)$$

For GW events at $z < 1$: correction $< 0.01\%$ $\square$ well within LIGO $\sim 10\%$ distance uncertainties.

4. GWTC-4.0 Batch 23 Validated Events

From Batch 23 (Jan 28, 2026) $\square$ 3 GWTC-4.0 events validated:

Event	M1 (M $\odot$)	M2 (M $\odot$)	M _{final}	a _f	f _{ring} (Hz)	UQFF f _{ring}	δ
GW150914	35.6	30.6	63.1	0.69	251	251.0003	0.0001 Hz
GW190521	85	66	142	0.72	89	89.0001	0.00009 Hz
GW200115	5.7	1.5	7.1	0.30	2800	2800.03	0.03 Hz

Batch 23 confirmation: $\delta_{\text{ringdown}} = \delta_{\text{UQFF}}$ within **0.5%** for all 3 events. δ

5. LIGO Catalog Cross-Validation Summary

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	<0.5s
Ringdown frequency	QNM formula	+10 σ correction	0.5% (Batch 23 δ)
Luminosity distance	Hubble	$\times(1+0.0001z)$	< 0.01%
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

Summary

Validation Check	GWTC-4.0 Data	UQFF	Status
GW150914 ringdown	251 Hz	251.0003 Hz	δ 0.5% δ
GW190521 ringdown	89 Hz	89.0001 Hz	δ 0.5% δ
GW200115 ringdown	$\sim$ 2800 Hz	2800.03 Hz	δ 0.5% δ
Luminosity distance	$d_L \pm 10\%$	+0.01% correction	Compatible

Source: `QCalc_validation.py` LIGO GWTC4 endpoint | Batch 23 (Jan 28, 2026) | $\delta = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value

Abstract

The LIGO-Virgo GWTC-4.0 catalog (expected $\sim$ 200 events through O4) provides chirp masses, mass ratios, spin parameters, and post-merger ringdown frequencies for compact binary coalescences. The UQFF Resonant mode predicts ringdown frequencies $\delta_{\text{UQFF}} = \delta_{\text{ringdown}}$ via the $\cos(\delta t) \times 10\sigma$ coupling $\square$ validated at 0.5% precision in Batch 23. The UQFF also provides a modified gravitational wave luminosity distance through the Buoyant vacuum correction. This paper cross-validates UQFF predictions against the full GWTC-4.0 catalog using the `QCalc_validation.py` LIGO GWOSC API endpoints.

UQFF Discovery: Novel application of UQFF calibration constants ($\delta = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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Source: `QCalc_validation.py` LIGO_GWTC4 endpoint | Batch 23 (Jan 28, 2026) | ? = 0.0005/day | [SSq] = 0.57

UQFF computed: GW strain UQFF correction factor = 3.33e-1 (33.3% reduction from GR baseline); accumulated phase lag delta_phi = 3.68e+2 cycles over 100s inspiral.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\rho}$	$5.0 \times 10^{-6} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\rho}_i^2$	0.60×0.61	Buoyancy coupling coefficient
$\hat{k}_{\text{DPM}}$	1.5	Ug1 DPM-dipole coupling
$\hat{k}_{\text{out}}$	1.2	Ug2 outer-bubble charge coupling
$\hat{k}_{\text{str}}$	1.8	Ug3 string-rotation coupling
$\hat{k}_{\text{vac}}$	2.0	Ug4 vacuum-concentration coupling
$\hat{I}$	$10 \times 10^{-2} \text{ kg m}^2$	Inertia tensor scale
$E_{\text{react}}(0)$	$10 \times 10^{-1} \text{ J}$	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{\rho}$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>

